

RATAN SINGH

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SUMMARY

BCA (AI & ML) student who builds full-stack and applied-AI systems end to end — MERN applications with JWT-secured REST APIs, FastAPI services for LLM and vision-language inference, and Retrieval-Augmented Generation pipelines — developed in version-controlled GitHub workflows.

EDUCATION

Bachelor of Computer Applications (BCA) — Artificial Intelligence & Machine Learning

Baba Farid Group of Institutions, Punjab, India | 2024 – 2027 | CGPA: 7.26/10 (through Semester 3)

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning

TECHNICAL SKILLS

Languages Python, JavaScript (ES6+), SQL

AI & ML LLM APIs, Retrieval-Augmented Generation (RAG), Tool Calling, Vector Embeddings, NumPy, Matplotlib

Backend FastAPI, Node.js, Express.js, REST APIs, JWT Authentication

Frontend React.js, HTML5, CSS3, Responsive UI

Databases MongoDB, Mongoose

DevOps Docker, Git, GitHub, Linux, Bash, CI/CD, Kubernetes, Jenkins, AWS (EC2, S3)

PROJECTS

SIMI AI — AI DevOps Assistant | <https://github.com/RataNBumbraH/SIMI-AI.git>

Python · FastAPI · React.js · LLM APIs · RAG · Docker · Kubernetes

- Built an AI-assisted DevOps platform that analyzes code repositories, application logs, and deployment issues, served through a FastAPI backend with a React.js conversational frontend.
- Implemented LLM tool calling with structured output schemas so the assistant executes scoped DevOps operations and returns parseable, actionable results instead of free-form text.
- Integrated Retrieval-Augmented Generation over technical documentation and project knowledge bases, grounding responses in repository-specific context rather than model priors.

SatQuery AI — Multimodal Satellite Change Analysis | <https://github.com/RataNBumbraH/Sat-Query-AI.git>

Python · LLaVA · LoRA · FastAPI · React.js · Docker

- Built a multimodal vision-language system that compares multi-temporal satellite imagery and identifies environmental and structural change across date periods.
- Adapted a LLaVA vision-language model using LoRA and served inference via FastAPI, handling image-payload processing, vector embeddings, and spatial query workflows.
- Implemented visual question-answering and context-grounding workflows that generate textual summaries of the temporal shifts detected across image pairs.

Village Information Hub — Full-Stack Community Platform | <https://github.com/RataNBumbraH/Website-For-Villages.git>

React.js · Node.js · Express.js · MongoDB · Mongoose · JWT

- Built a MERN platform managing village records, community camps, service requests, and citizen feedback through role-based workflows and administrative dashboards.
- Developed REST APIs in Node.js and Express with JWT-protected routes and role-based middleware separating administrator and citizen access.
- Modeled Mongoose schemas across users, service requests, and feedback logs to maintain data integrity and support CRUD operations from the React dashboard.

ACHIEVEMENTS & TRAINING

- Zinnovatio 3.0 Hackathon — built and demoed a working prototype with team Byte Eagles under a fixed time constraint.
- Data Analytics Training, Excellence Technology — data cleaning, NumPy, Matplotlib, and analytical processing workflows.
- Cybersecurity & Ethical Hacking Workshop — 5-day hands-on workshop on core offensive and defensive security concepts.